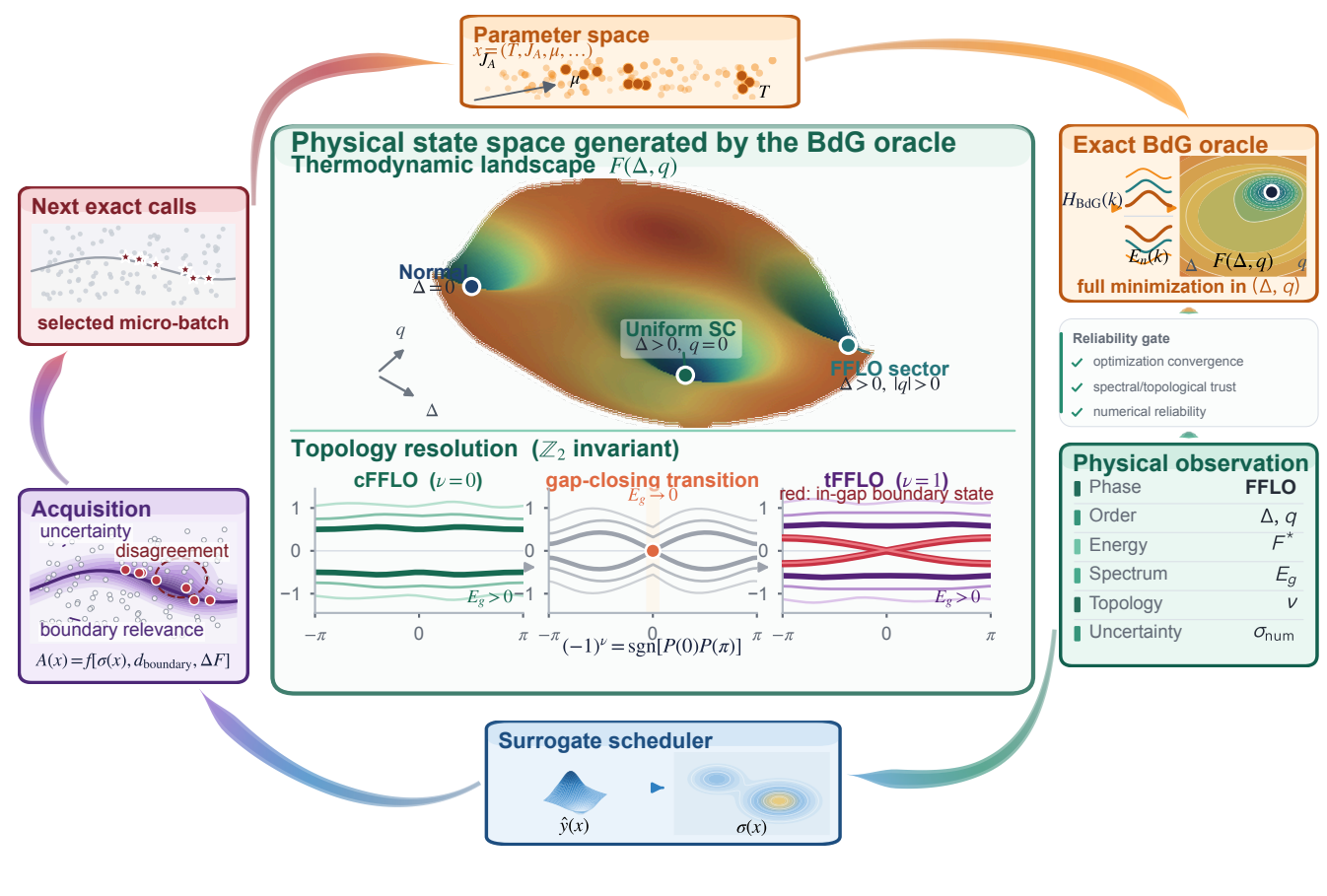

Parameter space
$x = (T, J_A, \mu, \ldots)$
μ
T
Physical state space generated by the BdG oracle
Thermodynamic landscape $F(\Delta, q)$
Normal
$\Delta = 0$
Uniform SC
$\Delta > 0,\ q = 0$
FFLO sector
$\Delta > 0,\ |q| > 0$
q
Δ
Topology resolution ($\mathbb{Z}_2$ invariant)
cFFLO ($\nu = 0$)
gap-closing transition
$E_g \to 0$
tFFLO ($\nu = 1$)
red: in-gap boundary state
$E_g > 0$
$E_g > 0$
$-\pi$
0
π
$(-1)^{\nu} = \mathrm{sgn}[P(0)P(\pi)]$
Exact BdG oracle
$H_{\mathrm{BdG}}(k)$
$E_n(k)$
$F(\Delta, q)$
full minimization in (Δ, q)
Reliability gate
optimization convergence
spectral/topological trust
numerical reliability
Physical observation
Phase FFLO
Order Δ, q
Energy F^*
Spectrum E_g
Topology ν
Uncertainty σ_{num}
Next exact calls
selected micro-batch
Acquisition
uncertainty
disagreement
boundary relevance
$A(x) = f[\sigma(x), d_{\mathrm{boundary}}, \Delta F]$
Surrogate scheduler
$\hat{y}(x)$
$\sigma(x)$